

Sangbeom Park

AI RESEARCH ENGINEER

Seoul, South Korea

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Work Experience

ATLO

CTO, Full-time

Seoul, S.Korea

Jun. 2024 – Apr. 2025

- Built a multimodal AI agent with emotion-aware dialogue generation, deployed on robot, iOS, and web platforms

NAVER Cloud

Machine Learning Engineer, Intern

Gyeonggi, S.Korea

Mar. 2024 – Jun. 2024

- Developed NewsBot, a personalized news delivery system powered by HyperCLOVA X

NAVER LABS

Machine Learning Engineer, Intern

Gyeonggi, S.Korea

Mar. 2023 – Sep. 2023

- Implemented unsupervised skill learning to develop safe and diverse quadruped locomotion behaviors (Mentor: Taeyoon Lee)

Education

Korea University

M.S. in Artificial Intelligence

Seoul, S.Korea

Sep. 2021 – Feb. 2025

- Thesis: Learning Diverse Manipulation Skills for Effective Semi-Autonomous Teleoperation
- Cumulative GPA: 4.42/4.5 | Credit: 40
- Supervisor: Sungjoon Choi

Dongguk University

B.S. in Electrical and Electronics Engineering, Minor in Intelligent Robotics Engineering

Seoul, S.Korea

Mar. 2015 – Aug. 2021

- EE GPA: 4.09/4.5 | IR GPA: 4.43/4.5 | Credit: 149
- Leave of absence for military service in KATUSA: Nov. 2016 – Aug. 2018

Publications

Quality-Diversity based Semi-Autonomous Teleoperation using Reinforcement Learning

Neural Networks

Sangbeom Park, Taerim Yoon, Joonhyung Lee, Sunghyun Park, and Sungjoon Choi

Nov. 2024

- [Project Page](#) | [Paper](#) | [Code](#) | [Poster](#)
- Published in JCR Top 10% Journal in Computer Science, Artificial Intelligence & Neurosciences (*Imapct Factor*: 7.8)
- Presented on Physical Human-Robot Interaction at *ICRA 2024 Workshop*

Visual Preference Inference: An Image Sequence-Based Preference Reasoning in Tabletop Object Manipulation

IROS

Joonhyung Lee, Sangbeom Park, Yongin Kwon, Jemin Lee, Sungjoon Choi

Oct. 2024

- [Project Page](#) | [Paper](#) | [Code](#) | [Video](#) | [Poster](#)
- Presented on Vision-Language Models for Navigation and Manipulation at *ICRA 2024 Workshop*

SPOTS: Stable Placement of Objects with Reasoning in Semi-Autonomous Teleoperation Systems

ICRA

Joonhyung Lee, Sangbeom Park, Jeongeun Park, Kyungjae Lee, and Sungjoon Choi

May. 2024

- [Project Page](#) | [Paper](#) | [Code](#) | [Video](#)

CLARA: Classifying and Disambiguating User Commands for Reliable Interactive Robotic Agents

RA-L

Jeongeun Park, Seungwon Lim, Joonhyung Lee, Sangbeom Park, Minsuk Chang, Youngjae Yu, and Sungjoon Choi

Dec. 2023

- [Project Page](#) | [Paper](#) | [Code](#) | [Video](#)

Semi-Autonomous Teleoperation via Learning Non-Prehensile Manipulation Skills

ICRA

Sangbeom Park, Yoonbyung Chai, Sunghyun Park, Jeongeun Park, Kyungjae Lee and Sungjoon Choi

May. 2022

• [Paper](#) | [Code](#) | [Video](#)

Learning Latent Prior for Rapid Adaptation of Legged Robots to Unexpected Amputation

Under Review

Sunghyun Park, Yoonbyung Chai, Seungyup Ka, Hyeonseong Kim, **Sangbeom Park**, Kevin Gim, Joohyung Kim, and Sungjoon Choi

2025

Projects

Interactive Multimodal Agent for Emotion-Aware Dialogue

ATLO

AI Research Engineer & CTO

Jun. 2024 - Apr. 2025

- Launched and operated a web-based dialogue agent service Atto and iOS app Ditto.
- Designed an asynchronous multimodal interaction pipeline.
- Implemented a RAG system using LangChain and LLMs with prompt engineering.
- Built full-stack web and iOS apps enabling multimodal chat with the agent.
- Secured 110M KRW seed funding at a 1.1B KRW valuation from Strong Ventures.
- Achieved 54,000+ conversation sessions in four months.

AI News Curation System

Naver Cloud

Machine Learning Engineer & Intern

Mar. 2024 - Jun. 2024

- Built backend infrastructure with AWS, Docker, and MySQL.
- Generated labeled data using LLMs and fine-tuned HyperCLOVA X for classification.
- Applied embedding similarity and reranking to filter and prioritize articles.
- Deployed as a chatbot, delivering curated news to 900+ employees.

Safe and Low-Noise Locomotion Skill Learning

NAVER LABS

Machine Learning Engineer & Intern

Mar. 2023 - Sep. 2023

- Designed a Safe Critic Network to suppress unsafe/noisy gaits.
- Used Hierarchical RL to learn diverse walking behaviors with safety-aware policies.
- Improved sim-to-real transfer via domain randomization in Isaac Gym.
- Developed a real-time Python-C++ control pipeline using the LCM communication framework.
- Reduced walking noise by 24%, validated in simulation and real hardware.

Automation Maintenance using Mobile Manipulator

SAMSUNG

M.S. Research Project

Mar. 2021 - Sep. 2023

- Created simulation environments in Unity and Gazebo for autonomous maintenance tasks.
- Developed navigation and manipulation capabilities for mobile manipulators in factory settings.

Presentations

Physical Human-Robot Interaction

Yokohama, Japan

Conference Workshop

May, 2024

- Poster presentation at 2024 IEEE International Conference on Robotics and Automation (ICRA)

Learning X in Physics-based Simulators

Seoul, S.Korea

Tech Talk

Mar. 2024

- Invited talk at CINAMON, AI Research Group

Shared-Autonomy for Robotics

Gyeonggi, S.Korea

Tech Talk

Feb. 2023

- Invited talk at NAVER LABS, Robot Dynamics & Control Group

Telerobotics

Conference Presentation

Philadelphia, U.S.A

May, 2022

- Oral presentation at 2022 IEEE International Conference on Robotics and Automation (ICRA)

Patent

2023 **10-2023-0125345**, Semi-Autonomous Teleoperation via Learning Manipulation

Korea

Skills

Programming	Python, C++, C#, Java
Physics Simulation	Isaac Sim, MuJoCo, Unity, Gazebo
Realworld Robots	<u>UR5e</u> , <u>Mini-Cheetah</u> , <u>Franka Panda</u> , <u>Pioneer 3-DX</u> , <u>Xycar</u>
DevOps	Amazon ECS, Docker
Languages	Korean (Native), English (Fluent)